

Advanced (Habanero)

- Q1.** What condition must a function satisfy to have an inverse?
- (A) The function must be linear
(B) The function must be quadratic
(C) The function must be one-to-one
(D) The function must be periodic
(E) The function must be continuous
- Q2.** Find the inverse of $f(x) = 2x + 1$.
- (A) $f^{-1}(x) = \frac{x-1}{2}$
(B) $f^{-1}(x) = \frac{x+1}{2}$
(C) $f^{-1}(x) = 2x - 1$
(D) $f^{-1}(x) = \frac{x}{2} - 1$
(E) $f^{-1}(x) = \frac{x}{2} + 1$
- Q3.** Which of the following functions has an inverse?
- (A) $f(x) = x^2$
(B) $f(x) = x^3$
(C) $f(x) = |x|$
(D) $f(x) = \sin(x)$
(E) $f(x) = x^4$
- Q4.** If $f(x) = x^2$ and $g(x) = \sqrt{x}$, what is $f(g(x))$?
- (A) x
(B) x^2
(C) $\sqrt{x}$
(D) x^4
(E) x^3
- Q5.** Verify that $f(x) = 3x - 2$ and $g(x) = \frac{x+2}{3}$ are inverses.
- (A) $f(g(x)) = x$ and $g(f(x)) = x$
(B) $f(g(x)) = x$ but $g(f(x)) \neq x$
(C) $g(f(x)) = x$ but $f(g(x)) \neq x$
(D) $f(g(x)) \neq x$ and $g(f(x)) \neq x$
(E) $f(g(x)) = g(f(x)) = x^2$
- Q6.** The inverse of $f(x) = \frac{1}{x}$ is
- (A) $f^{-1}(x) = \frac{1}{x}$
(B) $f^{-1}(x) = x$
(C) $f^{-1}(x) = \frac{1}{x^2}$
(D) $f^{-1}(x) = x^2$
(E) $f^{-1}(x) = \sqrt{x}$
- Q7.** Find the inverse of $f(x) = x^3 + 1$.
- (A) $f^{-1}(x) = \sqrt[3]{x-1}$
(B) $f^{-1}(x) = \sqrt[3]{x+1}$
(C) $f^{-1}(x) = x^3 - 1$
(D) $f^{-1}(x) = \frac{x}{3}$
(E) $f^{-1}(x) = x^2$
- Q8.** If a function f has an inverse, then f must be
- (A) one-to-one and onto
(B) only one-to-one
(C) only onto
(D) periodic
(E) continuous

Open Ended Questions (FRQ)

- Q9.** Find the inverse of the function $f(x) = 2x - 5$. Verify your answer using composition.
- Q10.** A company's profit P (in millions of dollars) is related to the amount x (in millions of dollars) invested. The relationship is given by $P(x) = 2x - 3$. Find the inverse of $P(x)$ and interpret its meaning.

Chef's Tips

- Check if the student verifies the inverse using composition for questions 9 and 10
- Ensure the student provides the correct interpretation of the inverse for question 10
- Watch for common algebraic errors when finding inverses